Machine Learning Methods for Predicting Professional Soccer Financial Trends

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1 Introduction

In the competitive world of professional football, player market values play a critical role in determining team composition, financial planning, and transfer strategies. Accurate predictions of these values can influence major decisions in club management, from signing new players to setting transfer prices and negotiating contracts. Despite the vast amount of data available, predicting player market values remains a complex challenge due to the multitude of factors that contribute to a player's worth.

Accurate market value and wage predictions, especially before and after transfer markets, can provide football clubs a competitive edge. By using robust predictive models, clubs can make more informed decisions, leading to better resource allocation, improved team performance, and potentially higher profitability. Additionally, understanding the impact of player transfers on both individual and club market values can offer strategic insights that enhance team management, such as starting squads and substitutions.

We address two key subtasks in this project:
1. Predicting player market-value and wage from player performance. The inputs to this algorithm will be aggregate 21 player features, both taken from the FIFA datasets and those that are engineered. The list of these statistics are provided in the sections below. We then use a jointly-trained, multi-task neural network (NN), called FIFANet, to output (1) the predicted wage and (2) predicted market value, in Euros.

2. Predicting player and club market value increases after transfer. To predict player market value increase, the inputs to this algorithm will be player statistics (e.g. 32 features like goals, assists, etc) over time. To predict club market value increase, the inputs to this algorithm will be team statistics (e.g. 89 features like pass completion, possession percentage, etc) over time. We then use a novel meta-model called ClubNet, which aggregates long short-term memory (LSTM), Random Forest, and Gradient Boosting Regressor, to predict the increase in (1) the market value of the player and (2) the market value of the club, in Euros.

2 Related Work

Several studies have explored the prediction of football player market values and wages using various methodologies. Chincholikar, et al used machine learning techniques to predict market values based on player attributes, but were limited by a specific time frame dataset [1]. Liu, et al. built a model-free reinforcement learning approach that builds an action-value function for player performance, but does not utilize that action-value function to make predictions about player market value or wage (instead about some proxy statistic) [2]. Schisano, et al predicted the future trajectory of players' market values, performance ratings, and wages using an RNN and LSTM but achieved poor accuracy relative to our ClubNet [3]. He, et al. developed models incorporating advanced performance metrics to predict transfer fees, showing improved accuracy but not addressing overall market values or changes post-transfer [4]. Chalupa, et al analyzed data from multiple sources to predict future market values of players in top European leagues, but was limited to the top 5 leagues [5]. Our research develops a comprehensive model that predicts both market values and wages, as well as the impact of transfers on player and club valuations, integrating thorough *international* data across a wide variety of clubs and skill levels.

3 Dataset and Features

To tackle the two subtasks, we used different datasets, each with different features well-suited to our sub-tasks. For Task 1, we used the FIFA dataset, offering detailed player attributes like height, weight, skill ratings (e.g., long shots and vision) and overall potential [11]. For task 2, we utilized a Transfermarket-derived dataset. This dataset includes detailed historical data on player market values, affiliations with various clubs, and temporal performance metrics such as goals, assists, minutes played, and disciplinary records, all recorded over several appearances/games [10]. More details about our exploratory data analysis can be found in the Appendix, see Figures 3, 4.

4 Methods

In this section, we will discuss the methods used in this project.

4.1 Feature Engineering, Selection, and Processing

For task 1, we compared three dimensionality reduction techniques: PCA, Gini impurity-based feature selection, and no dimensionality reduction. PCA, as discussed in class, reduces dimensionality by preserving variance. For Gini impurity-based feature selection, we constructed decision trees using Gini impurity to measure feature importance, selecting the top 21 features. The Gini impurity for a feature j across all splits is given by: $\text{Gini}(j) = 1 - \sum_{k=1}^{K} p_k^2$ where p_k is the probability of class k in the split. We pick the top 21 features that reduce the Gini impurity reduction for a split the most.

For task 2, we applied PCA to joint player-club data, reducing the player features from 83 to 32 and the club features from 132 to 89. During feature engineering, we handled missing values by dropping instances with missing target values and imputing missing feature values using the mean. All features were standardized using `StandardScaler`, and the target variable was normalized with `MinMaxScaler` to a range between 0 and 1. For our temporal models (LSTM baseline and ClubNet), we combed through the data to create sequences for players (goals, assists, etc over multiple games) and clubs (goals, no. of substitutions, etc over multiple games and seasons). Additional insights about the top PCA and their loadings (i.e., the main features they are comprised of) can be found in the Appendix, see Figure 5.

4.2 Baselines

For both tasks, we evaluated the performance of several baseline models, including **Ridge Regression**, which applies L_2-norm regularization to prevent overfitting while retaining all predictors; **Random Forest Regression**, which averages predictions from T decision trees, $\hat{y}(\mathbf{x}) = \frac{1}{T}\sum_{t=1}^{T} \hat{y}^{(t)}(\mathbf{x})$; **Decision Tree Regression**, which models non-linear interactions by splitting data into subsets; and **Gradient Boosting Regression (GBR)**, which sequentially builds trees to correct previous errors, optimizing a loss function: $\hat{y}_m(\mathbf{x}) = \hat{y}_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x})$, where $h_m(\mathbf{x})$ is the m-th tree and η is the learning rate. For Task 2, we have an additional **long short-term memory (LSTM)** baseline. LSTMs are a type of recurrent neural network (RNN) designed to capture long-term dependencies in sequential data by maintaining a memory cell that stores information over time. At each time step, the LSTM updates its memory and hidden state through gates (input, forget, and output), with the core update equation being: $h_t = o_t \cdot \tanh(C_t)$ where h_t is the hidden state, o_t is the output gate, and C_t is the memory cell state at time t. We trained all baselines using Sci-kit Learn and PyTorch on NVIDIA GPU via Google Cloud [6,7].

4.3 Task 1: FIFANet, A Multi-task NN

We used a multi-task neural network (MTNN) to predict both the wage and market value of professional soccer players. The MTNN shares two fully connected layers to exploit task-relatedness, with task-specific layers for individual predictions. The input layer consists of player attributes, and the shared layers include two fully connected layers with ReLU activations and dropout. The wage and market value predictions are computed using separate branches, each with two fully connected layers. The model is trained using a combined loss function, where the total loss is a weighted sum of the individual losses for wage and market value predictions. AdamW with weight decay is used for optimization. By sharing the initial layers, the MTNN leverages the relatedness of the tasks, improving the overall predictive performance. We evaluate MSE on the test set. We implemented FIFANet in PyTorch, and trained on an NVIDIA GPU via Google Cloud [7].

4.4 Task 2: ClubNet, Our Meta-Model Approach

We used a stacking, meta-model approach to enhance predictive accuracy by combining the strengths of multiple models: LSTM, Random Forest and GBR. As discussed above, the LSTM is fed the slice of the input with temporal data, such as sequences of player or club performance over time. Random Forest and GBR are fed the slice of the input with static features like player stamina, height, weight, and age. Predictions from the LSTM, Random Forest, and Gradient Boosting models are aggregated and used as input features for a meta-model, implemented as a Linear Regression. This meta-model learns to optimally weight the predictions from each base model. We use 4-fold cross-validation to compute test MSE and MAE. We trained ClubNet on PyTorch on an NVIDIA GPU [7].

5 Experiments, Results, and Discussion

We used mean squared error (MSE), defined as $\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ (where $\hat{y}$ is our prediction and y is our true value) as our main metric when training and evaluating all our models. We also report mean absolute error (MAE), defined as $\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$, and R-squared correlation coefficients: $R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$.

5.1 Task 1 Results

5.1.1 Training Experiments and Grid-Search for FIFANet

We use an 80-20 train-test split, with 56,923 examples in the train-set. We trained with MSE cost function. We ran a grid-search to find the optimal hyperparameters for FIFANet. The optimal parameters were: `batch_size` = 32, `dropout_rate` = 0.3, `shared_layers_dim` = (256, 128), `lr` = 0.001, `num_epochs` = 200, and `weight_decay` = 0.0001. The optimal training curves are shown in Figure 1a.

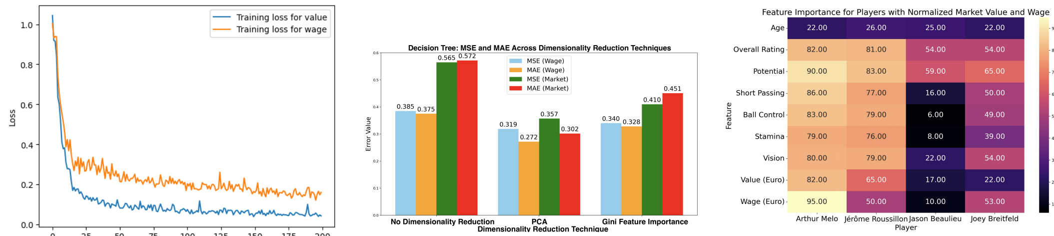
5.2 Results and Discussion

We compare the performance of our baselines with our custom jointly-trained NN. The results are shown in Table 3. For clarity, we have just shown the case with no dimensionality reduction; the full table of results can be found in the Appendix, Table 3.

Metric	Ridge Regression	Random Forest	Decision Tree	Gradient Boosting	FIFANet
MSE (Wage)	0.267	0.230	0.385	0.341	**0.092**
MAE (Wage)	0.239	0.194	0.375	0.314	**0.088**
R^2 (Wage)	0.75	0.78	0.70	0.80	**0.85**
MSE (Market)	0.335	0.280	0.565	0.325	**0.105**
MAE (Market)	0.340	0.230	0.572	0.317	**0.083**
R^2 (Market)	0.53	0.73	0.65	0.75	**0.80**

Table 1: Comparison of MSE, MAE, and R^2 scores for wage and market value predictions using no dimensionality reduction. Please see full results table in Appendix, Table 3.

First, the results in Table 1 demonstrate that FIFANet consistently outperformed all baseline models in predicting both wage and market value, achieving the highest R^2 scores alongside the lowest mean squared error (MSE) and mean absolute error (MAE). This makes sense with our intuition. The problem is clearly nonlinear and complex, with many important features. The baselines are



(a) FIFANet loss curves for player market value and wage prediction

(b) Impact of dimensionality reduction techniques on Decision Tree (baseline) accuracy

(c) Qualitative results showing cases where FIFANet outperformed *and* underperformed baselines

Figure 1: Task 1 Results

simple models that make assumptions about linearity (e.g. ridge regression) or hierarchical splits (e.g. Random Forest and Decision Tree). We believe that the reason why gradient boosting failed in this situation is due to overfitting, since it relies on an ensemble of decision trees to capture highly nonlinear data. The results in Figure 1b show the impact of dimensionality reduction techniques on decision tree accuracy. The other results are available in the Appendix, at Table 3. We can see that the lowest MSE and MAE for both tasks is achieved on the PCA dimensionality reduction. However, the Gini feature importance reduction still improves accuracy relative to no dimensionality reduction. This middling performance is due to the fact that Gini feature importance inherently has a hierarchy of features and therefore may miss important nonlinear aspects of the problem.

Qualitatively, FIFANet outperformed the baselines for players like *Arthur Melo* and *Jérôme Roussillon*, whose profiles are shown in Figure 1c, where its deep learning model captured important physiological features like stamina and vision. However, for players like *Jason Beaulieu* and *Joey Breitfeld*, with lower skill *and* physiological ratings, simpler models like Ridge Regression and Decision Tree Regression performed better since the problem was more straightforward——the players were less talented. These cases demonstrate that FIFANet excels with rich and complicated feature sets but struggles when the player profile has relatively homogeneous features (e.g., all poor ratings).

Finally, we mitigated overfitting by using regularization through weight decay. As discussed earlier, we found the optimal regularization parameter through a comprehensive grid search. We confirmed that overfitting had been mitigated by comparing the converged test and train loss, which were within 0.03 of each other.

5.3 Task 2 Results

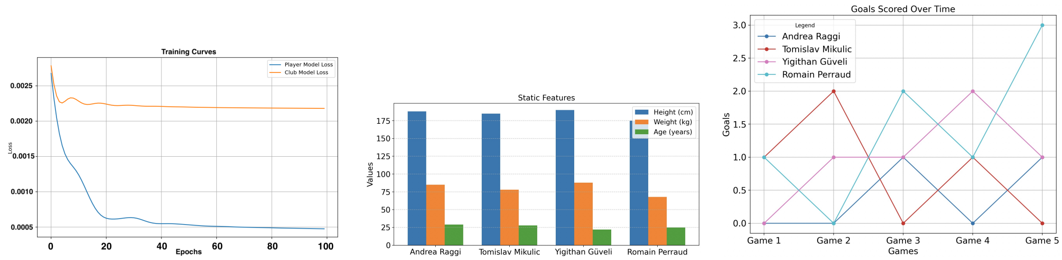
5.3.1 Training Experiments

To predict the player market value increase, we used a 4-fold cross-validation scheme. Each fold had 9,811 examples, each with 32 total features. We train using an MSE cost function (loss curves in Figure 2a), and report MSE results as averaged across all 4 folds. To predict the club market value increase, we used a similar 4-fold cross-validation MSE scheme, where each fold had 6,990 examples with 89 total features each.

5.3.2 Results and Discussion

Table 2: Performance Comparison for Task 2: Predicting Player and Club Market Value Increases (Normalized Data)

Model	Player MSE	Player MAE	Player R^2	Club MSE	Club MAE	Club R^2
Linear Regression	0.082	0.192	0.72	0.115	0.235	0.68
Decision Tree Regressor	0.065	0.174	0.80	0.092	0.208	0.76
Random Forest Regressor	0.054	0.162	0.86	0.075	0.184	0.83
Gradient Boosting Regressor	0.048	0.154	0.89	0.067	0.172	0.87
Long Short-Term Memory (LSTM)	0.0004	0.0088	0.7756	0.0022	0.0242	0.0490
ClubNet (Our NN)	**7.36e-5**	**0.0006**	**0.996**	**0.0022**	**0.0223**	**0.1412**



(a) ClubNet loss curves for player and club market value increase

(b) Static Feature Examples for 4 Players

(c) Temporal Feature Examples for 4 Players

Figure 2: Task 2 Results

The results in Table 2 show that ClubNet outperforms the simpler baseline models (such as linear regression, decision tree, random forest, and gradient boosting regressor) for predicting player and club market value increases, achieving the lowest MSE and MAE (0.0004 and 0.0088 for players; 0.0022 and 0.0223 for clubs). Linear Regression performs poorly due to its linear assumptions, while tree-based models like Decision Trees and Random Forests capture non-linearities but struggle with generalization. The base LSTM performs very well, and ClubNet offers significant improvements compared to it in player MSE/MAE, but minimal improvements in club MSE/MAE, even after a thorough grid-search for optimal hyperparameters. We believe ClubNet does much better than LSTM on the player task since it additionally incorporates static features via the GBR and Random Forest models. For the club MAE/MSE, we believe that the meta-model ClubNet suffers from redundancy, in that tree-based models might add unnecessary complexity to the sequential patterns that are well-captured by the LSTM—so, the benefits are limited compared to the base LSTM.

In terms of qualitative performance, ClubNet was able to capture nuances among both static and temporal features that simpler baselines were not able to. We show qualitative results for four players in Figure 2b and Figure 2c. For Andrea Raggi and Romain Perraud, ClubNet outperformed the baselines, likely due to its ability to capture temporal patterns in player upward trending goal performance. In contrast, baseline models like Random Forest and GBR might not have fully leveraged this temporal dependency, leading to less accurate predictions. On the other hand, Tomislav Mikulic and Yigithan Güveli showed better performance under the baseline models. These players displayed more fluctuating performance, but were younger and more agile, so their static features (e.g., height, weight, age) played a more significant role in predicting their market value increase. In these cases, the baseline models, which excel at handling static features, outperformed ClubNet.

Finally, we overcame overfitting challenges for the LSTM and regression-based submodels in ClubNet by using regularization, implemented via PyTorch weight decay [7]. We confirmed that overfitting had been mitigated by comparing the converged test/train loss, which were within 0.05 of each other.

6 Conclusion and Future Work

In this project, we have successfully demonstrated the effectiveness of advanced machine learning techniques, particularly multi-task neural networks (FIFANet), in predicting market values and wages of professional soccer players, as well as meta-models (ClubNet) to predict the potential market value increase of both players and clubs following transfers. By leveraging a combination of engineered features and dimensionality reduction techniques, we were able to develop models that outperform traditional regression models in terms of predictive accuracy, as shown by its superior performance on key metrics like MSE, MAE, and R-squared. To our knowledge, this is the first comprehensive work on these professional soccer financial tasks using multi-task NNs and meta-model NNs leveraging LSTMs with regression models. Our work also advances a new exploration of Gini-impurity-based dimensionality reduction as a computationally cheaper alternative to PCA; our mixed results are discussed above and are highlighted in more detail in our Appendix. Future work will focus on enhancing the models' generalizability. One avenue is to explore other external factors such as economic data, fan engagement statistics, and sentiment analysis in sports journalism. These datasets are larger and require more compute, which we can obtain using more credits on GCP.

References

[1] Chincholikar, S., Sadul, N., Thakkar, P., & Ashutosh. (2021). Predictive Analysis of Football Player Market Value Using Machine Learning Algorithms. International Research Journal of Modernization in Engineering Technology and Science, 3(6).

[2] Majewski, S., & Majewska, A. (2023). Determinants of football players' valuation: A systematic review. Journal of Economic Surveys, 37(2), 358-393.

[3] Schisano, L. (2022). Prediction of the soccer players market value. LUISS Guido Carli Thesis.

[4] He, Y. (2013). Predicting Market Value of Soccer Players Using Linear Modeling Techniques. UC Berkeley Thesis.

[5] Chalupa, V. (2024). Determinants of football players' market value during the season: Discrepancies between TOP 5 and the rest of the leagues. Charles University Thesis.

[6] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[7] Paszke, A., Gross, S., Chintala, S., Chanan, G., Yang, E., DeVito, Z., Lin, Z., Desmaison, A., Antiga, L., & Lerer, A. (2017). Automatic differentiation in PyTorch. In NIPS-W.

[8] Liu, G., Luo, Y., Schulte, O., & Kharrat, T. (2020). Deep soccer analytics: learning an action-value function for evaluating soccer players. Data Mining and Knowledge Discovery, 34(5), 1531-1559.

[9] Arrul, S. V., Subramanian, P., & Mafas, R. (2022). Predicting the football players' market value using neural network model: A data-driven approach. 2022 IEEE International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE), 1-7.

[10] Dcaribou. (n.d.). Dcaribou/transfermarkt-datasets: extract, prepare and publish Transfermarkt datasets. GitHub. https://github.com/dcaribou/transfermarkt-datasets.

[11] Leone, S. (2021, November 1). FIFA 22 complete player dataset. Kaggle. https://www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-player-dataset

Contributions

Akhilesh Balasingam: Exploratory Data Analysis, Task 1 Implementation (Dimensionality Reduction + FIFANet + Baselines), Report/Poster

Nael Ghoundale: Exploratory Data Analysis, Task 2 Implementation (Dimensionality Reduction + ClubNet + Baselines), Report/Poster

7 Appendix

7.1 Full Results, Task 1

Our full results for Task 1 are shown in Table 3.

Model	Metric	No Dimensionality Reduction	PCA	Gini Feature Importance
Ridge Regression	MSE (Wage)	0.267	0.260	0.251
	MAE (Wage)	0.239	0.210	0.205
	R^2 (Wage)	0.75	0.73	0.74
	MSE (Market)	0.335	0.310	0.303
	MAE (Market)	0.340	0.298	0.206
	R^2 (Market)	0.53	0.68	0.69
Random Forest	MSE (Wage)	0.230	0.227	0.245
	MAE (Wage)	0.194	0.178	0.230
	R^2 (Wage)	0.78	0.76	0.77
	MSE (Market)	0.280	0.273	0.275
	MAE (Market)	0.230	0.221	0.225
	R^2 (Market)	0.73	0.71	0.72
Decision Tree	MSE (Wage)	0.385	0.319	0.340
	MAE (Wage)	0.375	0.272	0.328
	R^2 (Wage)	0.70	0.68	0.69
	MSE (Market)	0.565	0.357	0.410
	MAE (Market)	0.572	0.302	0.451
	R^2 (Market)	0.65	0.63	0.64
Gradient Boosting	MSE (Wage)	0.341	0.211	0.209
	MAE (Wage)	0.314	0.163	0.151
	R^2 (Wage)	0.80	0.78	0.79
	MSE (Market)	0.325	0.267	0.253
	MAE (Market)	0.317	0.210	0.204
	R^2 (Market)	0.75	0.73	0.74
FIFANet	MSE (Wage)	**0.092**	0.159	0.163
	MAE (Wage)	**0.088**	0.173	0.174
	R^2 (Wage)	**0.85**	0.83	0.82
	MSE (Market)	**0.105**	0.151	0.182
	MAE (Market)	**0.083**	0.192	0.193
	R^2 (Market)	**0.80**	0.69	0.68

Table 3: Comparison of MSE, MAE, and R^2 scores for wage and market value predictions across baselines and FIFANet, with different dimensionality reduction techniques.

7.2 Data Visualization

Some examples of the work we did in exploratory data analysis are shown in Figures 3, 4.

7.3 PCA results on Transfermarket datasets

The results from our PCA dimensionality reduction are shown in Figure 5.

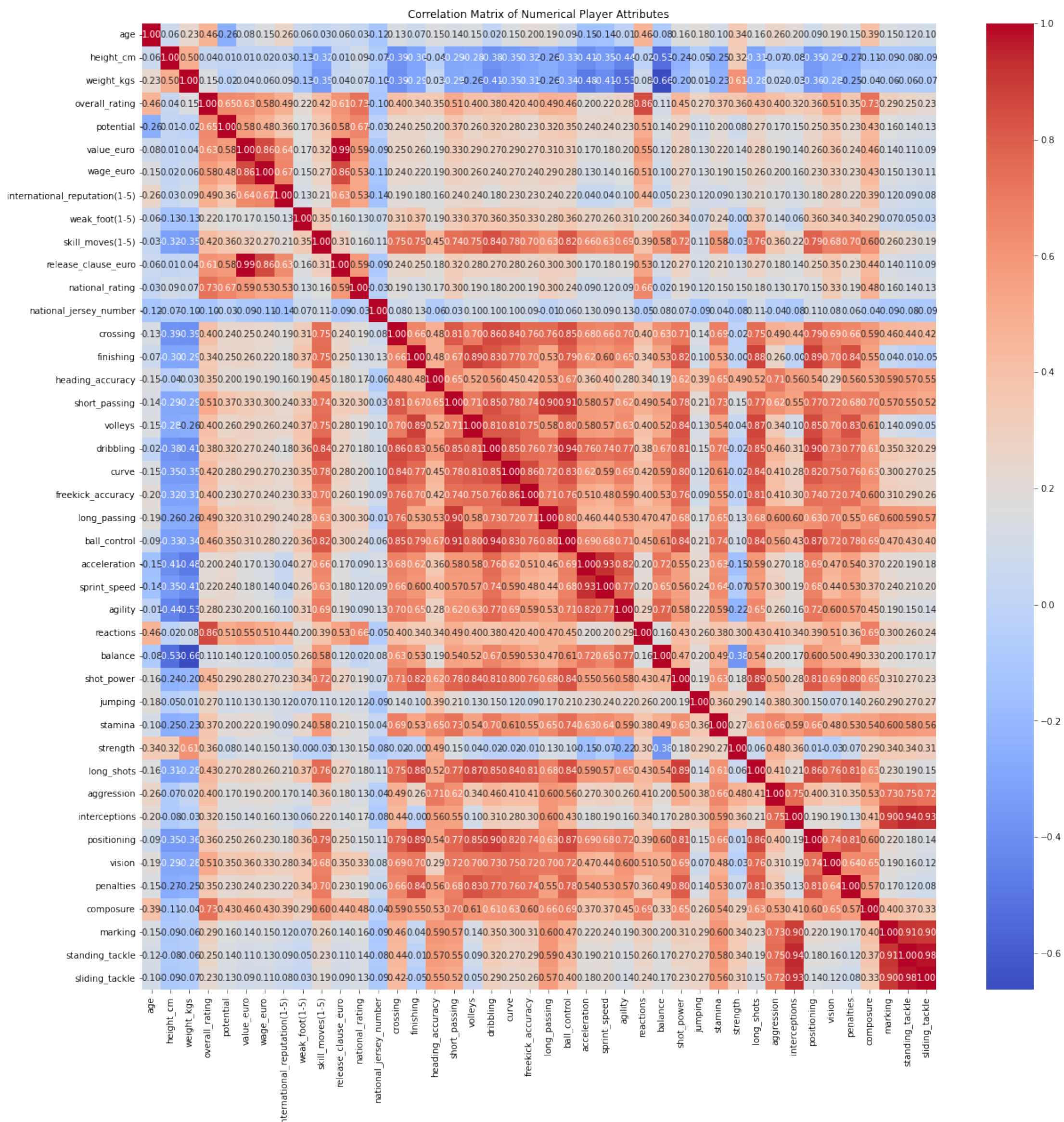


Figure 3: Correlation matrix of numerical player attributes (FIFA dataset)

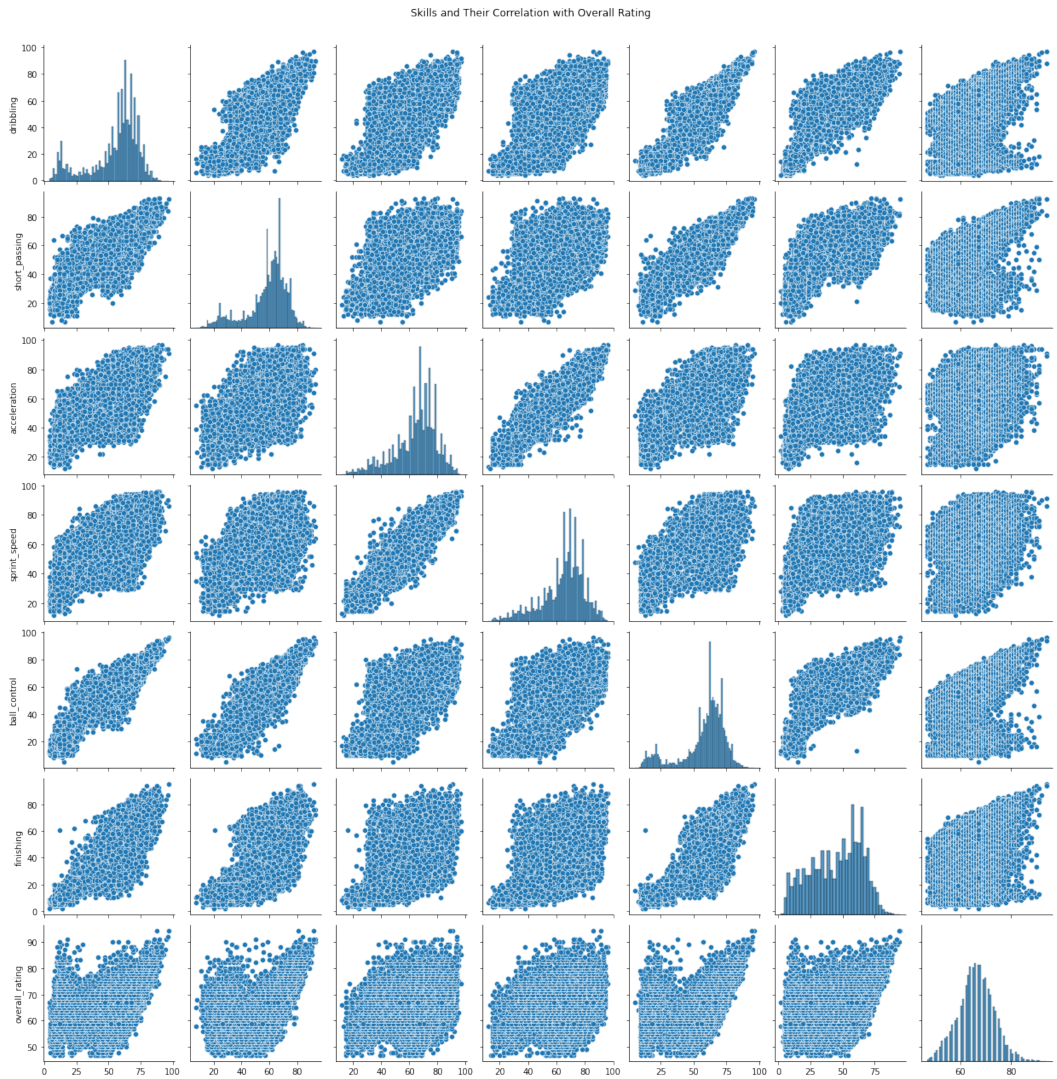
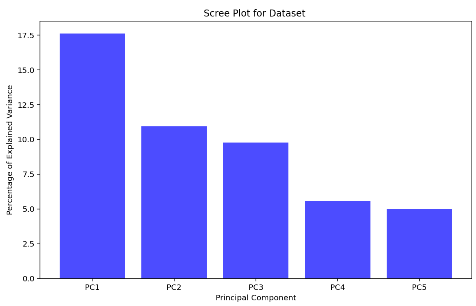
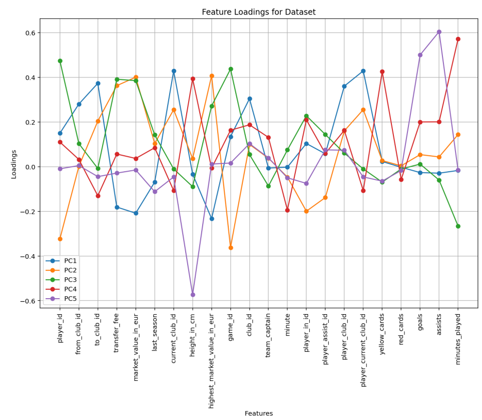


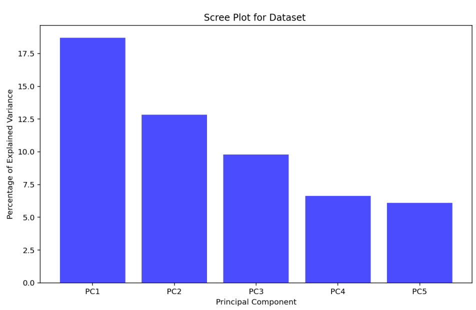
Figure 4: Skills and their correlation to overall ratings (FIFA dataset)



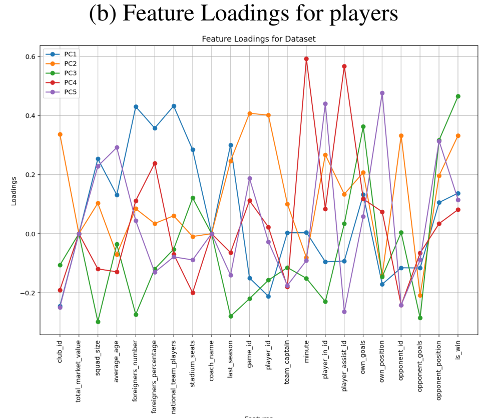
(a) Top 5 PC for players



(b) Feature Loadings for players



(c) Top 5 PC for clubs



(d) Feature Loadings for clubs

Figure 5: PCA Dimensionality Reduction from Transfermarket Dataset